

# **Module 4 – Kubernetes & Cloud**

## **1. Kubernetes**

### **What is Kubernetes?**

Kubernetes is a container orchestration platform used to manage, deploy, and scale containerized applications.

### **Why Kubernetes?**

- Automates deployment
- Manages container scaling
- Ensures high availability

### **Key Features**

- Auto-scaling
  - Load balancing
  - Self-healing
- 

## **2. Pods**

### **What is a Pod?**

A pod is the smallest deployable unit in Kubernetes that contains one or more containers.

### **Key Points**

- Runs containers together
  - Shares network and storage
  - Basic unit of deployment
- 

## **3. Scaling**

### **What is Scaling?**

Scaling means increasing or decreasing application resources based on demand.

### **Types of Scaling**

- Horizontal Scaling – Add more instances
- Vertical Scaling – Increase resources

### **Benefits**

- Handles high traffic
  - Improves performance
  - Ensures reliability
- 

## 4. Cloud Introduction

### What is Cloud Computing?

Cloud computing provides servers, storage, and services over the internet.

### Popular Cloud Providers

- AWS
- Google Cloud
- Microsoft Azure

### Benefits

- Scalable
  - Cost-effective
  - Accessible globally
- 

## 5. Deployment

### What is Deployment?

Deployment is the process of making an application live and accessible to users.

### Steps

- Build application
  - Containerize using Docker
  - Deploy on Kubernetes
  - Host on cloud
- 

## 6. Key Concepts to Remember

- Kubernetes manages containers
  - Pods are smallest deployment units
  - Scaling handles traffic
  - Cloud provides hosting
  - Deployment makes apps live
-

## 7. Summary

Kubernetes and cloud platforms are essential for modern DevOps workflows. They enable scalable, reliable, and efficient deployment of applications, ensuring high availability and performance in production environments.